# Hades Model Report

v0.8.4 GitHub

Hastie & Smith, et al. (2026)

**MODEL CONFIGURATION**

This report summarises the calibration of the model against the reference dataset, including parameter estimates and convergence diagnostics.

| | |
|---|---|
| Mode | **Alkaline** |
| Submode | **-** |
| Data suite | **./madeira_porto_santo.xlsx** |
| $FeO/Fe^{2+}$ | **0.9** |
| Stages Run | Primary Primitive Basic Intermediate Evolved |

## 1. Results at a Glance

All of the mineral abundances and fractional crystallisation (FC) values for each phase are shown in the table below, as well as the resultant total fractions for the performed amount of fractional crystallisation over all phases.

The total melt fraction crystallised across all phases is 89.61 wt.%

| | Primary | Primitive | Basic | Intermediate | Evolved | Total |
|---|---|---|---|---|---|---|
| **Olivine** | 99.32 | 31.05 | 11.44 | 8.36 | 1.49 | 32.3 |
| **Spinel** | 0.68 | 1.73 | | | | 1.12 |
| **Plagioclase** | | 9.63 | 52.79 | 59.96 | 49.98 | 21.56 |
| **K-feldspar** | | | | 2.71 | 35.79 | 1.83 |
| **Biotite** | | | | 1.96 | 9.03 | 0.6 |
| **Magnetite** | | 0.8 | 15.1 | 2.42 | 1.94 | 2.83 |
| **Ilmenite** | | 1.48 | 0.66 | 0.59 | 1.78 | 1.12 |
| **Titanaugite** | | 53.84 | 18.35 | 23.72 | | 37.5 |
| **Alkali-amphibole** | | 1.47 | 1.65 | 0.28 | | 1.14 |
| **FC%** | 10.0 | 60.0 | 33.0 | 41.0 | 27.0 | |

Table 1: Optimised mineral weights for the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

## 2. Parameter Configuration

The parameters below define the model as it was executed for this run.

| | Primary | Primitive | Basic | Intermediate | Evolved |
|---|---|---|---|---|---|
| **Plagioclase** | 1.2 | 1.2 | 1.2 | 1.2 | 1.2 |
| **Clinopyroxene** | 0.28 | 0.28 | 0.28 | 0.28 | 0.28 |
| **Orthopyroxene** | 0.29 | 0.29 | 0.29 | 0.29 | 0.29 |
| **Amphibole** | 0.37 | 0.37 | 0.37 | 0.37 | 0.37 |
| **K-feldspar** | 1.3 | 1.3 | 1.3 | 1.3 | 1.3 |
| **Biotite** | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 |
| **Titanaugite** | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 |
| **Nepheline** | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 |
| **Leucite** | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 |
| **Nosean/Hauyne** | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 |
| **Alkali-amphibole** | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 |

Table 2: Partition coefficients for the model phases.

## 3. Magma Evolution

### Initial Compositions

| | Sample | SiO2 | TiO2 | Al2O3 | Fe2O3 | MgO | CaO | Na2O | K2O |
|---|---|---|---|---|---|---|---|---|---|
| **Primary** | 4 | 43.72 | 2.38 | 11.91 | 12.96 | 15.5 | 10.79 | 2.1 | 0.64 |
| **Primitive** | 37 | 44.51 | 2.63 | 13.84 | 13.62 | 12.05 | 9.69 | 2.83 | 0.85 |
| **Basic** | 187 | 51.14 | 2.28 | 19.19 | 9.78 | 2.68 | 7.78 | 5.58 | 1.57 |
| **Intermediate** | 202 | 58.98 | 1.23 | 18.87 | 5.92 | 1.23 | 5.83 | 6.13 | 1.81 |
| **Evolved** | 218 | 67.8 | 0.54 | 17.45 | 2.81 | 0.37 | 1.49 | 5.96 | 3.58 |
| **End** | 222 | 71.14 | 0.2 | 16.12 | 1.35 | 0.2 | 1.01 | 6.65 | 3.33 |

Table 3: Major oxide compositions of the model phases (wt.%).

### Computed Compositions

| | SiO2 | TiO2 | Al2O3 | Fe2O3 | MgO | CaO | Na2O | K2O |
|---|---|---|---|---|---|---|---|---|
| **Primary** | 43.72 | 2.38 | 11.91 | 12.96 | 15.5 | 10.79 | 2.1 | 0.64 |
| **Primitive** | 44.19 | 2.63 | 13.19 | 12.95 | 12.05 | 11.94 | 2.33 | 0.71 |
| **Basic** | 51.51 | 2.25 | 19.17 | 9.8 | 2.68 | 7.82 | 5.06 | 1.7 |
| **Intermediate** | 57.54 | 1.22 | 19.09 | 5.99 | 1.21 | 5.95 | 6.55 | 2.46 |
| **Evolved** | 65.0 | 0.54 | 17.33 | 2.78 | 0.25 | 1.52 | 8.78 | 3.8 |
| **End** | 67.64 | 0.19 | 16.12 | 1.34 | 0.24 | 1.0 | 9.6 | 3.88 |

Table 4: Major oxide compositions of the model phases (wt.%).

# 4. Optimisation Results

## Primary > Primitive

PRIMARY

| | |
|---|---|
| Population size | 50 |
| Iterations | 460/500 |
| Cost function | 1.57 |
| Trace analysis? | False |

### Magma Inspection

| | Initial | Expected | Computed | % Error |
|---|---|---|---|---|
| **SiO2** | 43.72 | 44.51 | 44.19 | 0.71 |
| **TiO2** | 2.38 | 2.63 | 2.63 | 0.21 |
| **Al2O3** | 11.91 | 13.84 | 13.19 | 4.69 |
| **Fe2O3** | 12.96 | 13.62 | 12.95 | 4.89 |
| **MgO** | 15.5 | 12.05 | 12.05 | 0.0 |
| **CaO** | 10.79 | 9.69 | 11.94 | 23.28 |
| **Na2O** | 2.1 | 2.83 | 2.33 | 17.45 |
| **K2O** | 0.64 | 0.85 | 0.71 | 16.16 |

Table 5: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

### Optimised Parameters

| | Value |
|---|---|
| **FC%** | 10.0 |
| **Olivine** | 99.32 |
| **Spinel** | 0.68 |
| **Plagioclase** | 0.0 |
| **K-feldspar** | 0.0 |
| **Biotite** | 0.0 |
| **Magnetite** | 0.0 |
| **Ilmenite** | 0.0 |
| **Titanaugite** | 0.0 |
| **Nepheline** | 0.0 |
| **Leucite** | 0.0 |
| **Nosean/Hauyne** | 0.0 |
| **Alkali-amphibole** | 0.0 |

Table 6: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

### Bulk Crystallising Assemblage Composition

| | SiO2 | TiO2 | Al2O3 | Fe2O3 | MgO | CaO | Na2O | K2O |
|---|---|---|---|---|---|---|---|---|
| **wt.%** | 39.32 | 0.09 | 0.33 | 13.99 | 45.8 | 0.47 | 0.0 | 0.0 |

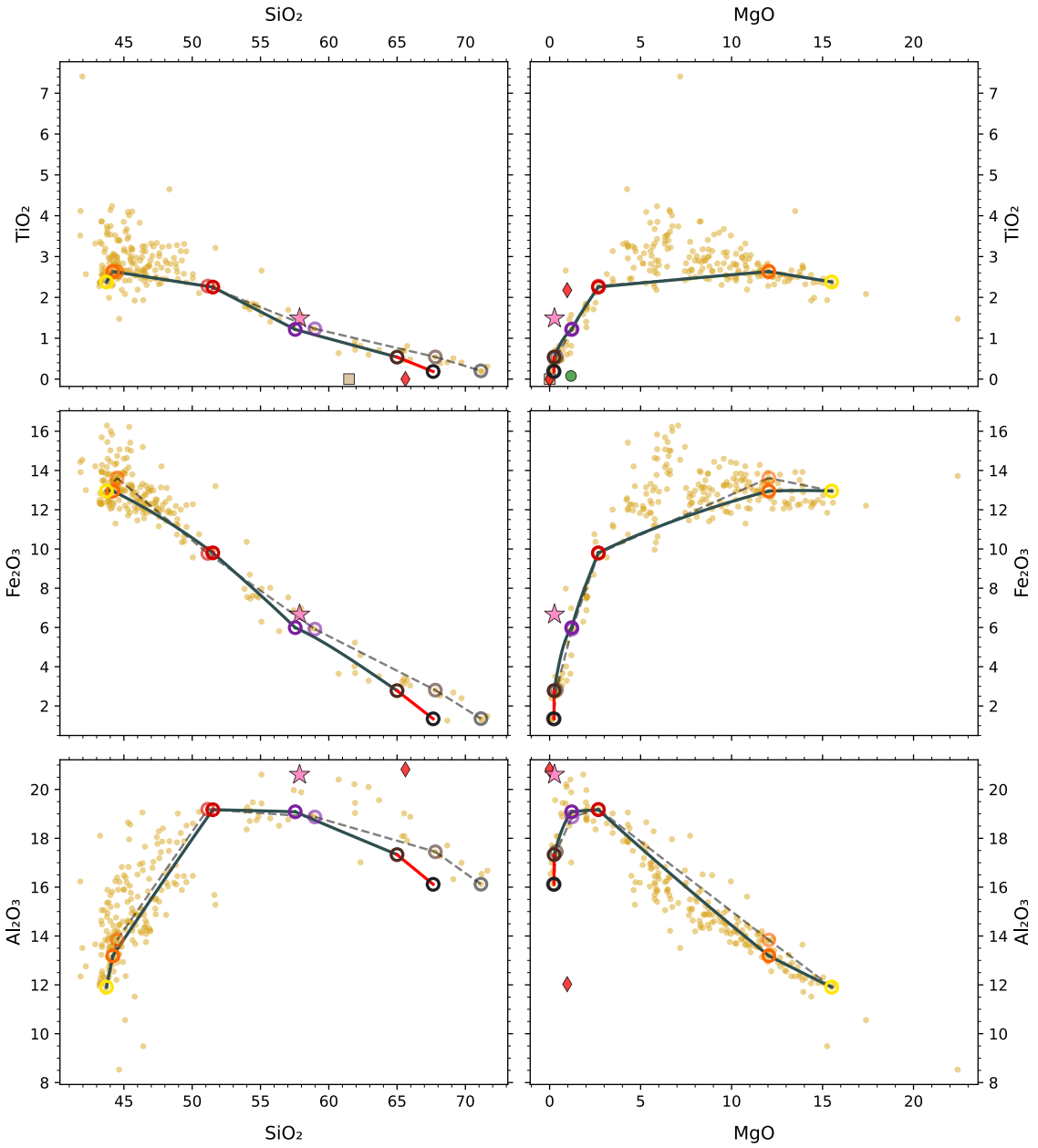
Table 7: Bulk mineralogy of the model phases (wt.%).

### Cotectic Compositions

| | SiO2 | TiO2 | Al2O3 | Fe2O3 | MgO | CaO | Na2O | K2O |
|---|---|---|---|---|---|---|---|---|
| **olivine** | 39.58 | 0.09 | 0.12 | 13.88 | 46.02 | 0.31 | 0.0 | 0.0 |
| **spinel** | 0.26 | 0.53 | 30.33 | 30.52 | 13.68 | 24.34 | 0.34 | 0.0 |
| **plagioclase** | | | | | | | | |
| **kfeldspar** | | | | | | | | |
| **biotite** | | | | | | | | |
| **magnetite** | | | | | | | | |
| **ilmenite** | | | | | | | | |
| **titanaugite** | | | | | | | | |
| **nepheline** | | | | | | | | |
| **leucite** | | | | | | | | |
| **nosean_hauyne** | | | | | | | | |
| **alkali_amphibole** | | | | | | | | |

Table 8: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

### Bivariate Plot




Bivariate plots continued overleaf.

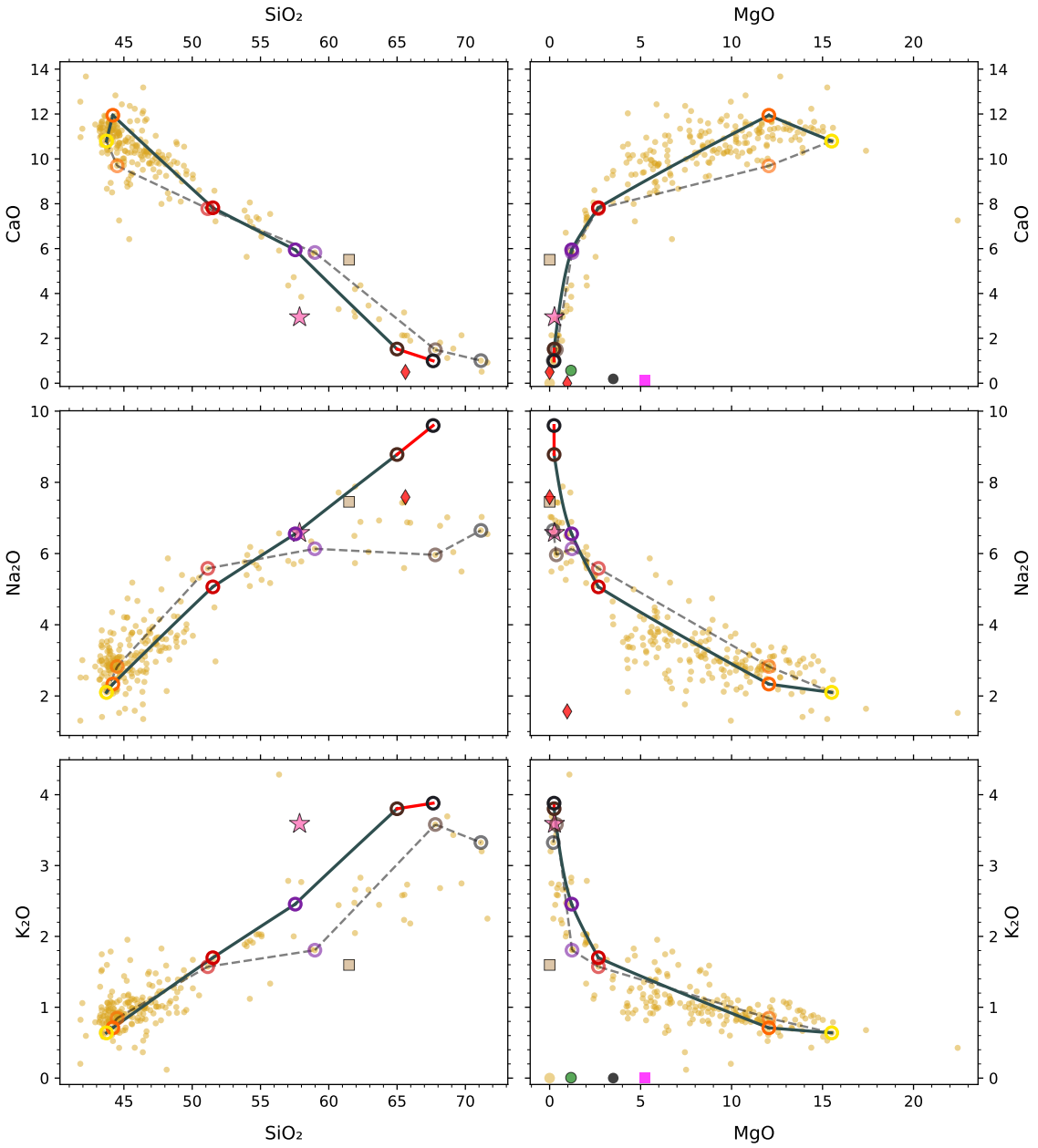



Figure 1: Bivariate plots of the data suite with the data-based and modelled LLODs. The Primary to Primitive transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

## Primitive > Basic

PRIMITIVE

| | |
|---|---|
| Population size | 50 |
| Iterations | 500/500 |
| Cost function | 0.42 |
| Trace analysis? | False |

### Magma Inspection

| | Initial | Expected | Computed | % Error |
|---|---|---|---|---|
| **SiO2** | 44.19 | 51.14 | 51.51 | 0.73 |
| **TiO2** | 2.63 | 2.28 | 2.25 | 1.08 |

| | Initial | Expected | Computed | % Error |
|---|---|---|---|---|
| **Al2O3** | 13.19 | 19.19 | 19.17 | 0.1 |
| **Fe2O3** | 12.95 | 9.78 | 9.8 | 0.28 |
| **MgO** | 12.05 | 2.68 | 2.68 | 0.05 |
| **CaO** | 11.94 | 7.78 | 7.82 | 0.51 |
| **Na2O** | 2.33 | 5.58 | 5.06 | 9.3 |
| **K2O** | 0.71 | 1.57 | 1.7 | 8.05 |

Table 9: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

### Optimised Parameters

| | Value |
|---|---|
| **FC%** | 60.0 |
| **Olivine** | 31.05 |
| **Spinel** | 1.73 |
| **Plagioclase** | 9.63 |
| **K-feldspar** | 0.0 |
| **Biotite** | 0.0 |
| **Magnetite** | 0.8 |
| **Ilmenite** | 1.48 |
| **Titanaugite** | 53.84 |
| **Nepheline** | 0.0 |
| **Leucite** | 0.0 |
| **Nosean/Hauyne** | 0.0 |
| **Alkali-amphibole** | 1.47 |

Table 10: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

### Bulk Crystallising Assemblage Composition

| | SiO2 | TiO2 | Al2O3 | Fe2O3 | MgO | CaO | Na2O | K2O |
|---|---|---|---|---|---|---|---|---|
| **wt.%** | 38.97 | 2.88 | 9.28 | 16.41 | 17.12 | 14.77 | 0.51 | 0.05 |

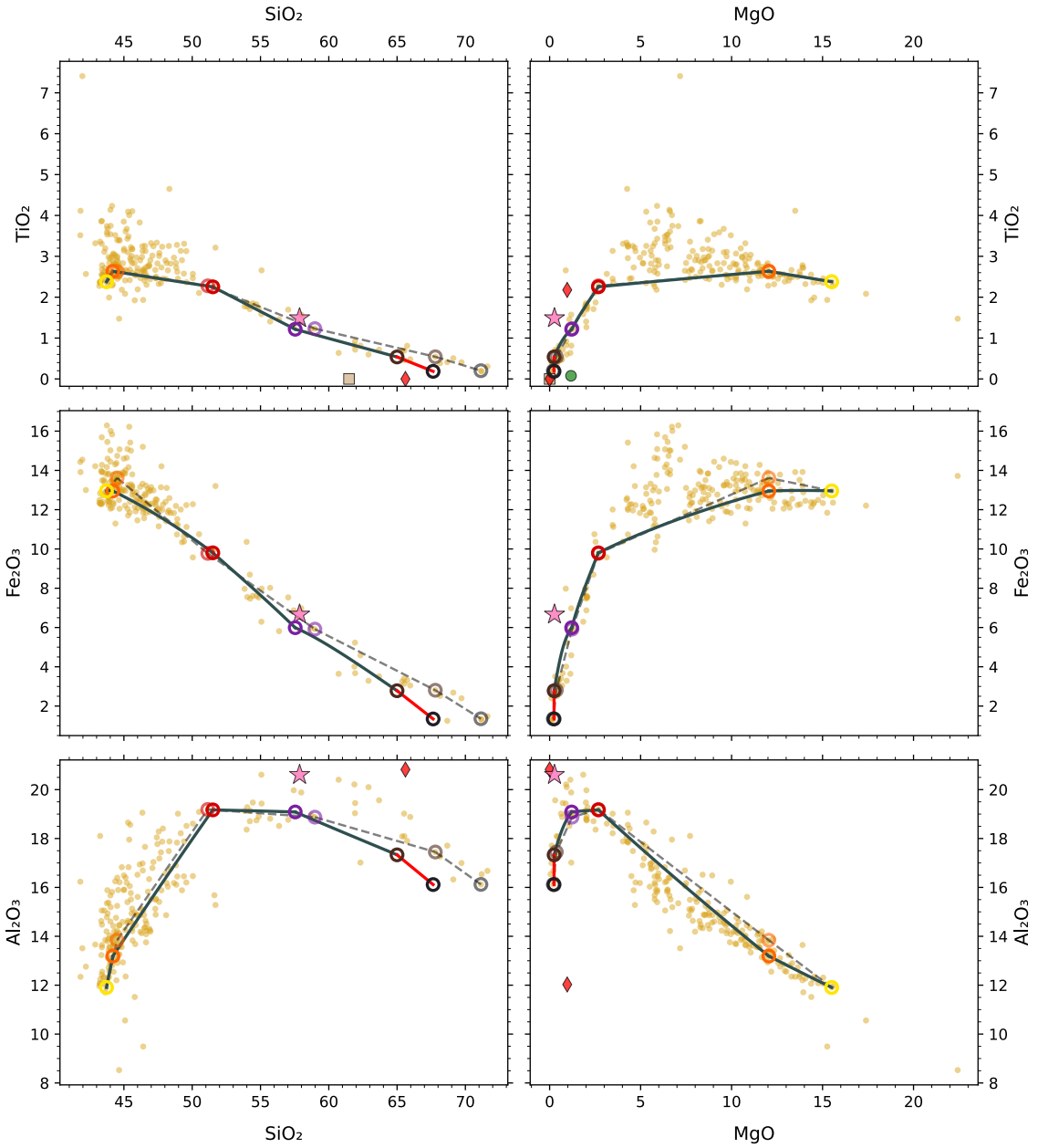
Table 11: Bulk mineralogy of the model phases (wt.%).

### Cotectic Compositions

| | SiO2 | TiO2 | Al2O3 | Fe2O3 | MgO | CaO | Na2O | K2O |
|---|---|---|---|---|---|---|---|---|
| **olivine** | 38.08 | 0.07 | 0.35 | 20.27 | 40.64 | 0.59 | 0.0 | 0.0 |
| **spinel** | 0.26 | 0.53 | 30.33 | 30.52 | 13.68 | 24.34 | 0.34 | 0.0 |
| **plagioclase** | 49.16 | 0.0 | 32.25 | 0.33 | 0.1 | 15.48 | 2.5 | 0.19 |
| **kfeldspar** | | | | | | | | |
| **biotite** | | | | | | | | |
| **magnetite** | 0.0 | 5.0 | 0.0 | 95.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| **ilmenite** | 0.0 | 50.0 | 0.0 | 50.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| **titanaugite** | 40.16 | 3.84 | 10.27 | 14.11 | 7.76 | 23.5 | 0.36 | 0.0 |
| **nepheline** | | | | | | | | |
| **leucite** | | | | | | | | |
| **nosean_hauyne** | | | | | | | | |
| **alkali_amphibole** | 53.62 | 0.46 | 0.62 | 31.55 | 5.19 | 1.36 | 5.05 | 2.15 |

Table 12: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

**Bivariate Plot**




Bivariate plots continued overleaf.

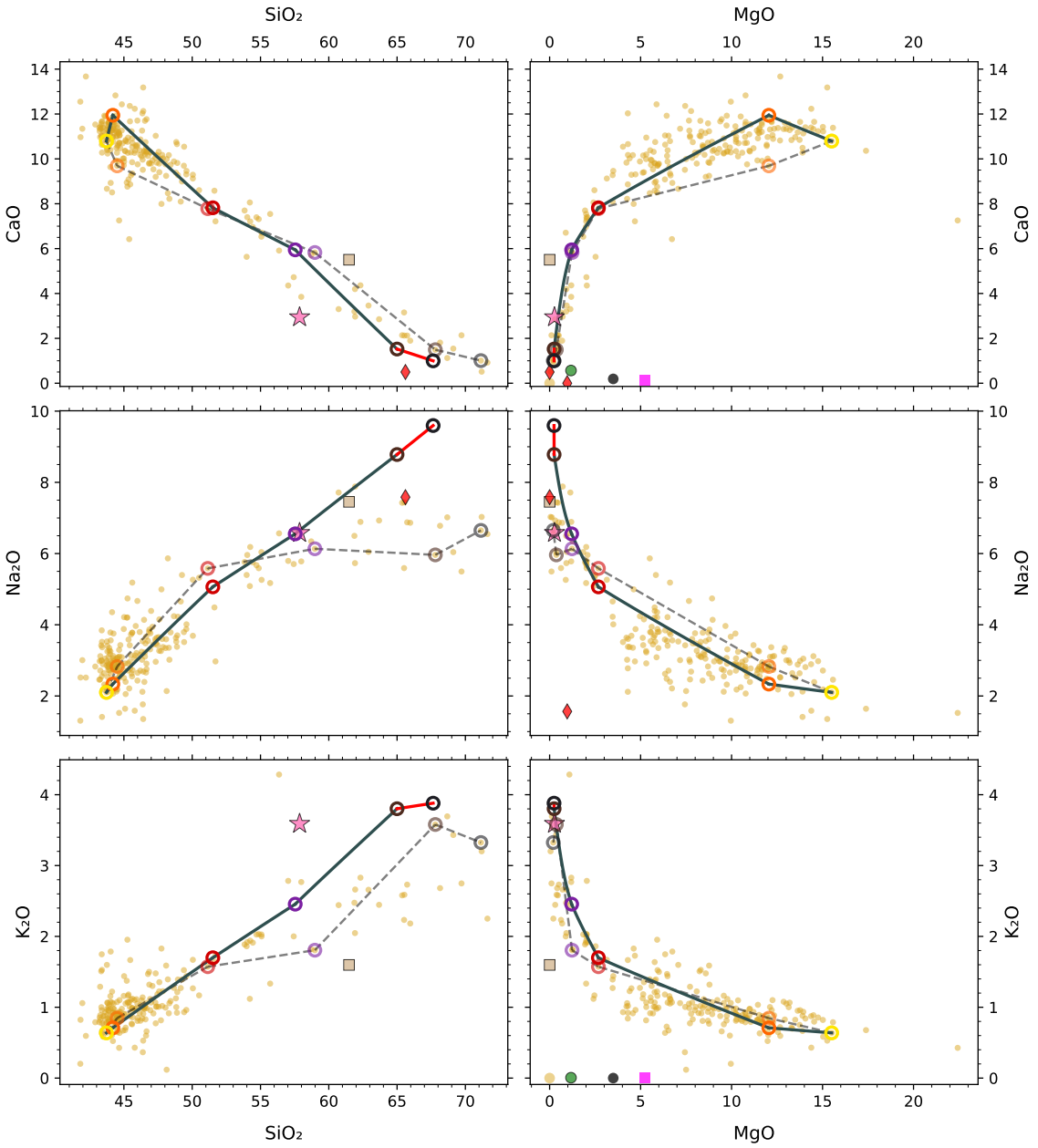



Figure 2: Bivariate plots of the data suite with the data-based and modelled LLODs. The Primitive to Basic transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

## Basic > Intermediate

BASIC

| | |
|---|---|
| Population size | 50 |
| Iterations | 500/500 |
| Cost function | 1.00 |
| Trace analysis? | False |

**Magma Inspection**

| | Initial | Expected | Computed | % Error |
|---|---|---|---|---|
| **SiO2** | 51.51 | 58.98 | 57.54 | 2.44 |
| **TiO2** | 2.25 | 1.23 | 1.22 | 1.23 |

| | Initial | Expected | Computed | % Error |
|---|---|---|---|---|
| **Al2O3** | 19.17 | 18.87 | 19.09 | 1.12 |
| **Fe2O3** | 9.8 | 5.92 | 5.99 | 1.24 |
| **MgO** | 2.68 | 1.23 | 1.21 | 1.64 |
| **CaO** | 7.82 | 5.83 | 5.95 | 2.04 |
| **Na2O** | 5.06 | 6.13 | 6.55 | 6.84 |
| **K2O** | 1.7 | 1.81 | 2.46 | 35.99 |

Table 13: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

### Optimised Parameters

| | Value |
|---|---|
| **FC%** | 33.0 |
| **Olivine** | 11.44 |
| **Spinel** | 0.0 |
| **Plagioclase** | 52.79 |
| **K-feldspar** | 0.0 |
| **Biotite** | 0.0 |
| **Magnetite** | 15.1 |
| **Ilmenite** | 0.66 |
| **Titanaugite** | 18.35 |
| **Nepheline** | 0.0 |
| **Leucite** | 0.0 |
| **Nosean/Hauyne** | 0.0 |
| **Alkali-amphibole** | 1.65 |

Table 14: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

### Bulk Crystallising Assemblage Composition

| | SiO2 | TiO2 | Al2O3 | Fe2O3 | MgO | CaO | Na2O | K2O |
|---|---|---|---|---|---|---|---|---|
| **wt.%** | 39.09 | 4.36 | 19.38 | 18.24 | 5.07 | 11.65 | 2.04 | 0.16 |

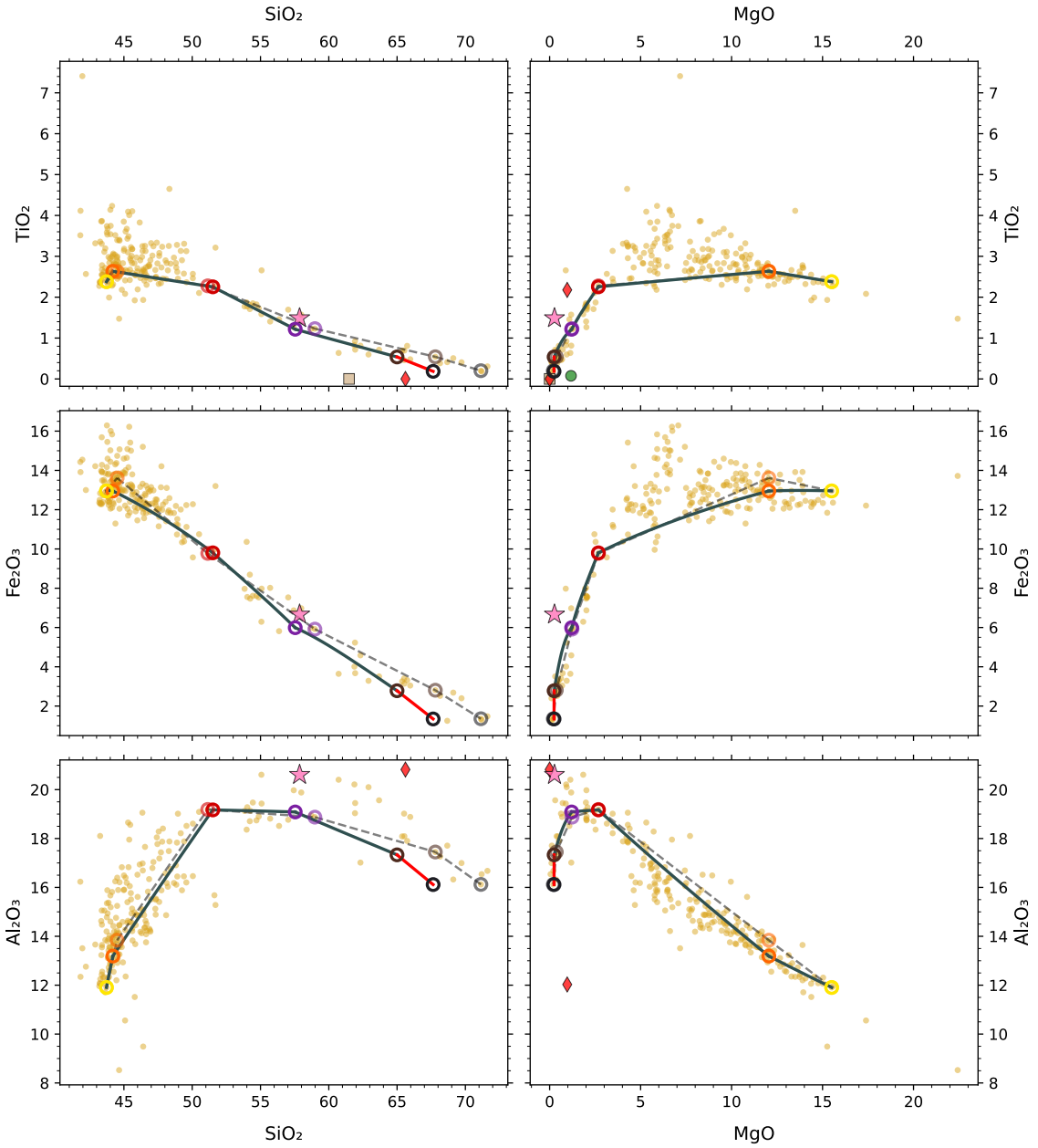
Table 15: Bulk mineralogy of the model phases (wt.%).

### Cotectic Compositions

| | SiO2 | TiO2 | Al2O3 | Fe2O3 | MgO | CaO | Na2O | K2O |
|---|---|---|---|---|---|---|---|---|
| **olivine** | 34.43 | 0.04 | 1.0 | 35.64 | 27.97 | 0.91 | 0.0 | 0.0 |
| **spinel** | | | | | | | | |
| **plagioclase** | 51.96 | 0.0 | 30.66 | 0.0 | 0.0 | 13.47 | 3.67 | 0.24 |
| **kfeldspar** | | | | | | | | |
| **biotite** | | | | | | | | |
| **magnetite** | 0.0 | 19.64 | 3.1 | 73.57 | 3.5 | 0.19 | 0.0 | 0.0 |
| **ilmenite** | 0.0 | 51.14 | 0.46 | 43.03 | 5.25 | 0.12 | 0.0 | 0.0 |
| **titanaugite** | 37.24 | 5.68 | 14.18 | 12.25 | 6.67 | 23.88 | 0.09 | 0.0 |
| **nepheline** | | | | | | | | |
| **leucite** | | | | | | | | |
| **nosean_hauyne** | | | | | | | | |
| **alkali_amphibole** | 53.62 | 0.46 | 0.62 | 31.55 | 5.19 | 1.36 | 5.05 | 2.15 |

Table 16: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

**Bivariate Plot**




Bivariate plots continued overleaf.

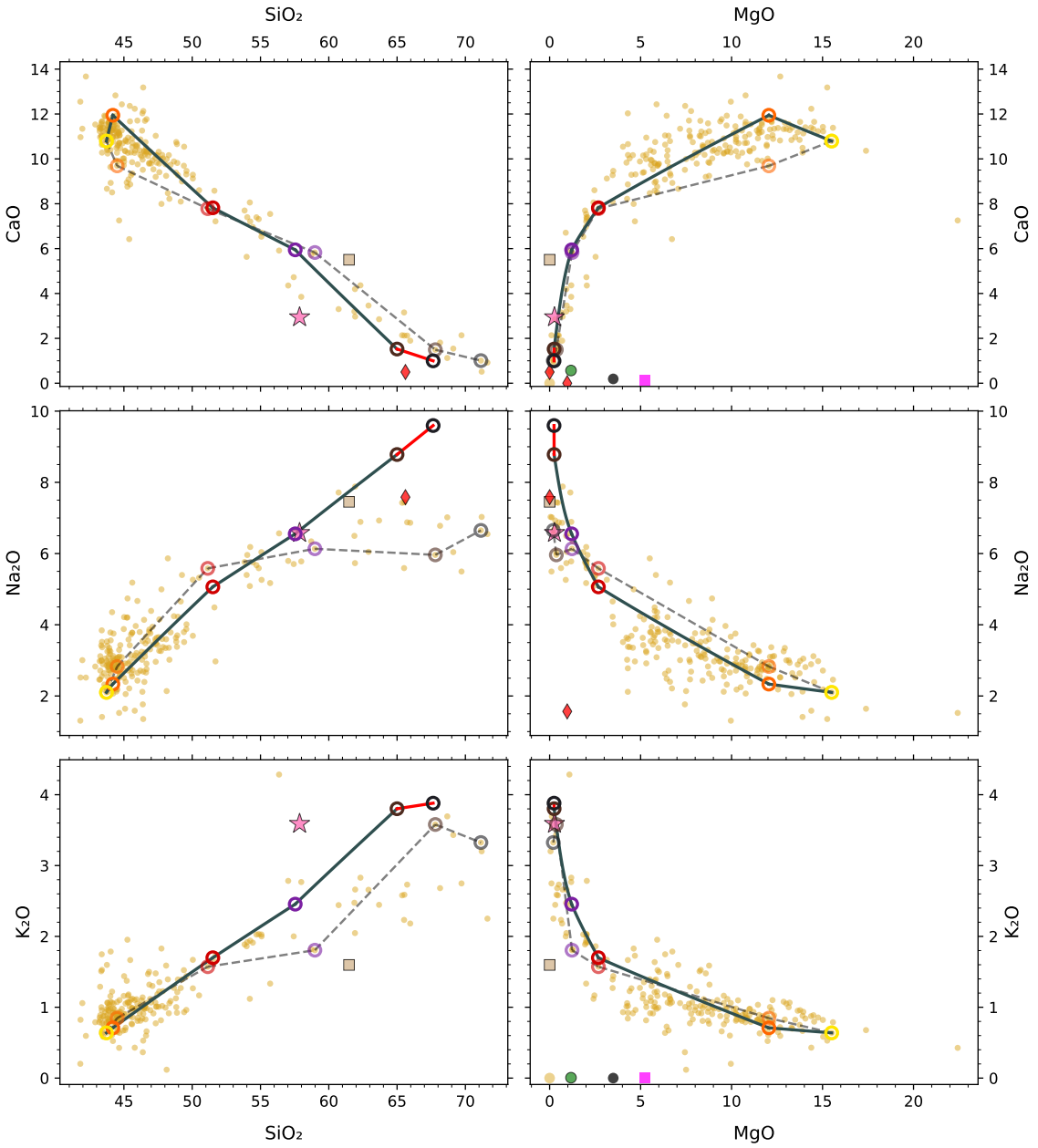



Figure 3: Bivariate plots of the data suite with the data-based and modelled LLODs. The Basic to Intermediate transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

## Intermediate > Evolved

INTERMEDIATE

| | |
|---|---|
| Population size | 50 |
| Iterations | 500/500 |
| Cost function | 1.89 |
| Trace analysis? | False |

### Magma Inspection

| | Initial | Expected | Computed | % Error |
|---|---|---|---|---|
| **SiO2** | 57.54 | 67.8 | 65.0 | 4.13 |
| **TiO2** | 1.22 | 0.54 | 0.54 | 0.79 |

| | Initial | Expected | Computed | % Error |
|---|---|---|---|---|
| **Al2O3** | 19.09 | 17.45 | 17.33 | 0.67 |
| **Fe2O3** | 5.99 | 2.81 | 2.78 | 0.94 |
| **MgO** | 1.21 | 0.37 | 0.25 | 33.06 |
| **CaO** | 5.95 | 1.49 | 1.52 | 2.02 |
| **Na2O** | 6.55 | 5.96 | 8.78 | 47.28 |
| **K2O** | 2.46 | 3.58 | 3.8 | 6.21 |

Table 17: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

### Optimised Parameters

| | Value |
|---|---|
| **FC%** | 41.0 |
| **Olivine** | 8.36 |
| **Spinel** | 0.0 |
| **Plagioclase** | 59.96 |
| **K-feldspar** | 2.71 |
| **Biotite** | 1.96 |
| **Magnetite** | 2.42 |
| **Ilmenite** | 0.59 |
| **Titanaugite** | 23.72 |
| **Nepheline** | 0.0 |
| **Leucite** | 0.0 |
| **Nosean/Hauyne** | 0.0 |
| **Alkali-amphibole** | 0.28 |

Table 18: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

### Bulk Crystallising Assemblage Composition

| | SiO2 | TiO2 | Al2O3 | Fe2O3 | MgO | CaO | Na2O | K2O |
|---|---|---|---|---|---|---|---|---|
| **wt.%** | 46.57 | 2.21 | 21.69 | 11.51 | 1.85 | 12.31 | 3.35 | 0.52 |

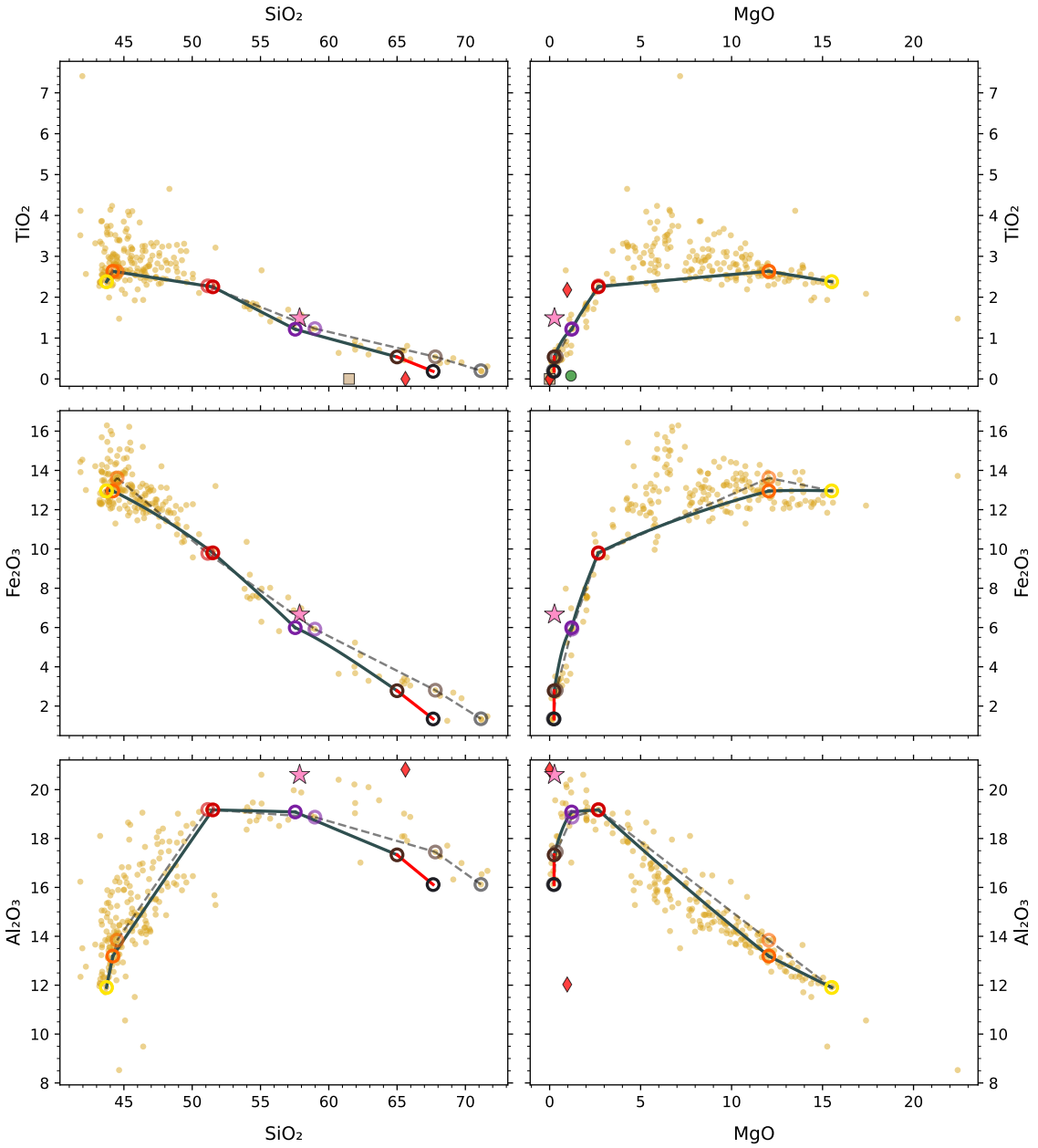
Table 19: Bulk mineralogy of the model phases (wt.%).

### Cotectic Compositions

| | SiO2 | TiO2 | Al2O3 | Fe2O3 | MgO | CaO | Na2O | K2O |
|---|---|---|---|---|---|---|---|---|
| **olivine** | 26.48 | 0.29 | 1.93 | 69.8 | 0.7 | 0.78 | 0.02 | 0.01 |
| **spinel** | | | | | | | | |
| **plagioclase** | 54.82 | 0.0 | 28.74 | 0.0 | 0.0 | 10.93 | 5.18 | 0.34 |
| **kfeldspar** | 65.61 | 0.0 | 20.82 | 0.0 | 0.0 | 0.5 | 7.58 | 5.49 |
| **biotite** | 36.69 | 3.1 | 14.41 | 33.16 | 4.18 | 0.17 | 0.15 | 8.14 |
| **magnetite** | 0.0 | 19.64 | 3.1 | 73.57 | 3.5 | 0.19 | 0.0 | 0.0 |
| **ilmenite** | 0.0 | 51.14 | 0.46 | 43.03 | 5.25 | 0.12 | 0.0 | 0.0 |
| **titanaugite** | 37.24 | 5.68 | 14.18 | 12.25 | 6.67 | 23.88 | 0.09 | 0.0 |
| **nepheline** | | | | | | | | |
| **leucite** | | | | | | | | |
| **nosean_hauyne** | | | | | | | | |
| **alkali_amphibole** | 53.62 | 0.46 | 0.62 | 31.55 | 5.19 | 1.36 | 5.05 | 2.15 |

Table 20: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

**Bivariate Plot**




Bivariate plots continued overleaf.

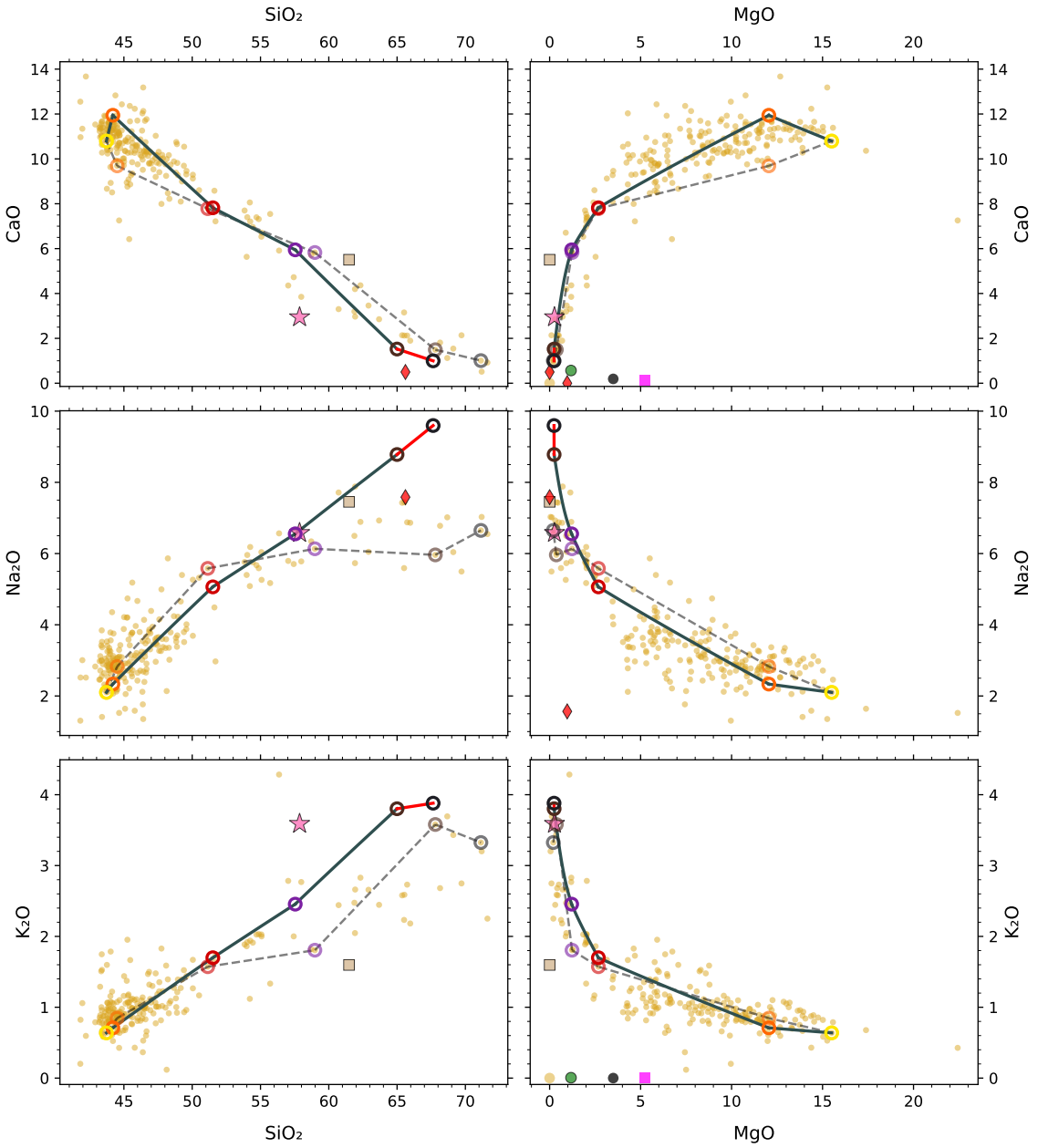



Figure 4: Bivariate plots of the data suite with the data-based and modelled LLODs. The Intermediate to Evolved transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

## Evolved > End

EVOLVED

| | |
|---|---|
| Population size | 50 |
| Iterations | 500/500 |
| Cost function | 2.01 |
| Trace analysis? | False |

**Magma Inspection**

| | Initial | Expected | Computed | % Error |
|---|---|---|---|---|
| **SiO2** | 65.0 | 71.14 | 67.64 | 4.92 |
| **TiO2** | 0.54 | 0.2 | 0.19 | 7.92 |

| | Initial | Expected | Computed | % Error |
|---|---|---|---|---|
| **Al2O3** | 17.33 | 16.12 | 16.12 | 0.04 |
| **Fe2O3** | 2.78 | 1.35 | 1.34 | 0.79 |
| **MgO** | 0.25 | 0.2 | 0.24 | 20.03 |
| **CaO** | 1.52 | 1.01 | 1.0 | 0.98 |
| **Na2O** | 8.78 | 6.65 | 9.6 | 44.29 |
| **K2O** | 3.8 | 3.33 | 3.88 | 16.7 |

Table 21: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

### Optimised Parameters

| | Value |
|---|---|
| **FC%** | 27.0 |
| **Olivine** | 1.49 |
| **Spinel** | 0.0 |
| **Plagioclase** | 49.98 |
| **K-feldspar** | 35.79 |
| **Biotite** | 9.03 |
| **Magnetite** | 1.94 |
| **Ilmenite** | 1.78 |
| **Titanaugite** | 0.0 |
| **Nepheline** | 0.0 |
| **Leucite** | 0.0 |
| **Nosean/Hauyne** | 0.0 |
| **Alkali-amphibole** | 0.0 |

Table 22: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

### Bulk Crystallising Assemblage Composition

| | SiO2 | TiO2 | Al2O3 | Fe2O3 | MgO | CaO | Na2O | K2O |
|---|---|---|---|---|---|---|---|---|
| **wt.%** | 57.86 | 1.49 | 20.61 | 6.66 | 0.27 | 2.94 | 6.58 | 3.59 |

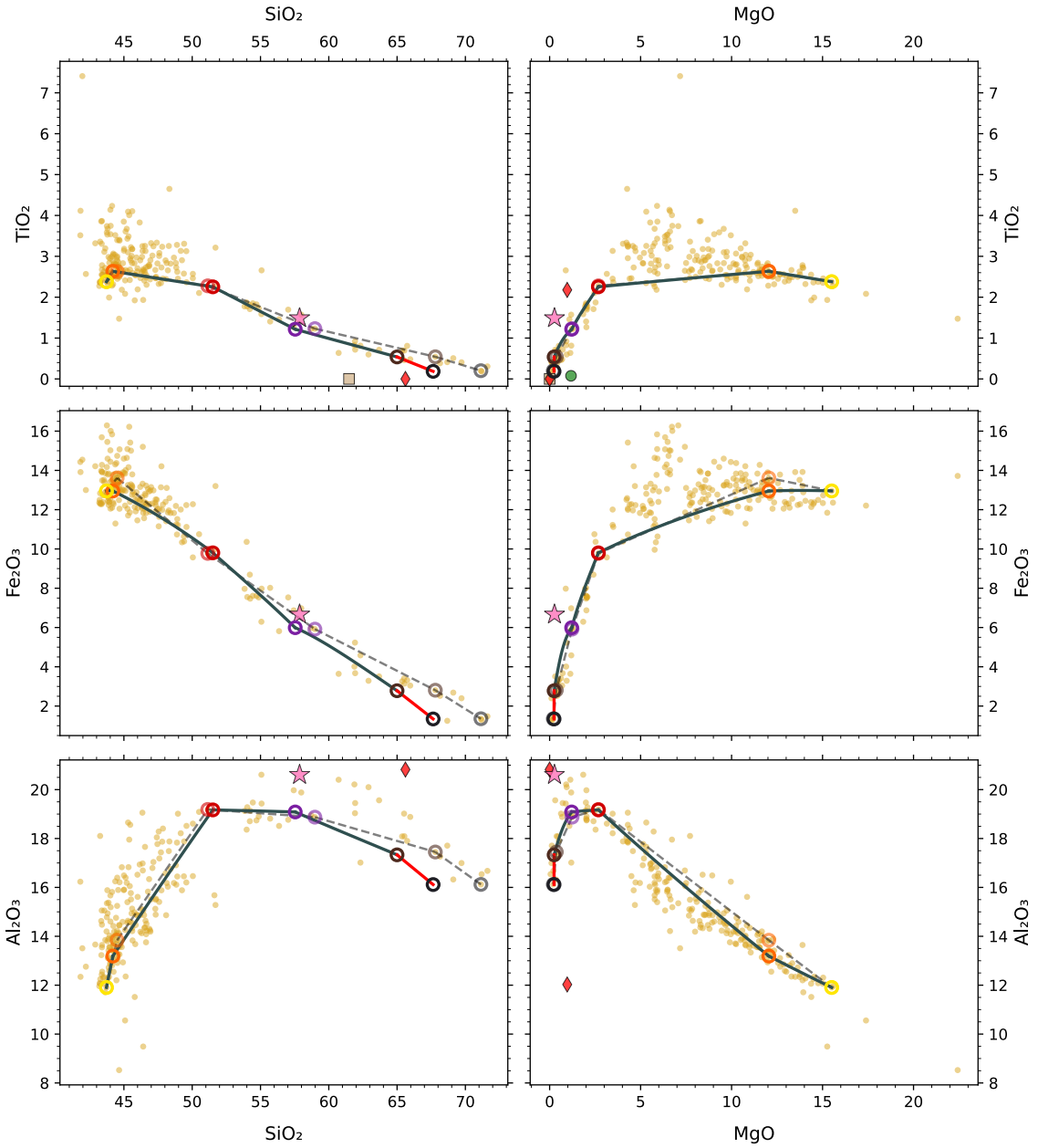
Table 23: Bulk mineralogy of the model phases (wt.%).

### Cotectic Compositions

| | SiO2 | TiO2 | Al2O3 | Fe2O3 | MgO | CaO | Na2O | K2O |
|---|---|---|---|---|---|---|---|---|
| **olivine** | 26.95 | 0.08 | 1.5 | 69.71 | 1.17 | 0.57 | 0.01 | 0.01 |
| **spinel** | | | | | | | | |
| **plagioclase** | 61.48 | 0.0 | 23.97 | 0.0 | 0.0 | 5.5 | 7.45 | 1.6 |
| **kfeldspar** | 65.61 | 0.0 | 20.82 | 0.0 | 0.0 | 0.5 | 7.58 | 5.49 |
| **biotite** | 36.02 | 2.18 | 12.03 | 38.06 | 0.97 | 0.0 | 1.57 | 9.18 |
| **magnetite** | 0.0 | 19.64 | 3.1 | 73.57 | 3.5 | 0.19 | 0.0 | 0.0 |
| **ilmenite** | 0.0 | 51.14 | 0.46 | 43.03 | 5.25 | 0.12 | 0.0 | 0.0 |
| **titanaugite** | | | | | | | | |
| **nepheline** | | | | | | | | |
| **leucite** | | | | | | | | |
| **nosean_hauyne** | | | | | | | | |
| **alkali_amphibole** | | | | | | | | |

Table 24: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

**Bivariate Plot**




Bivariate plots continued overleaf.

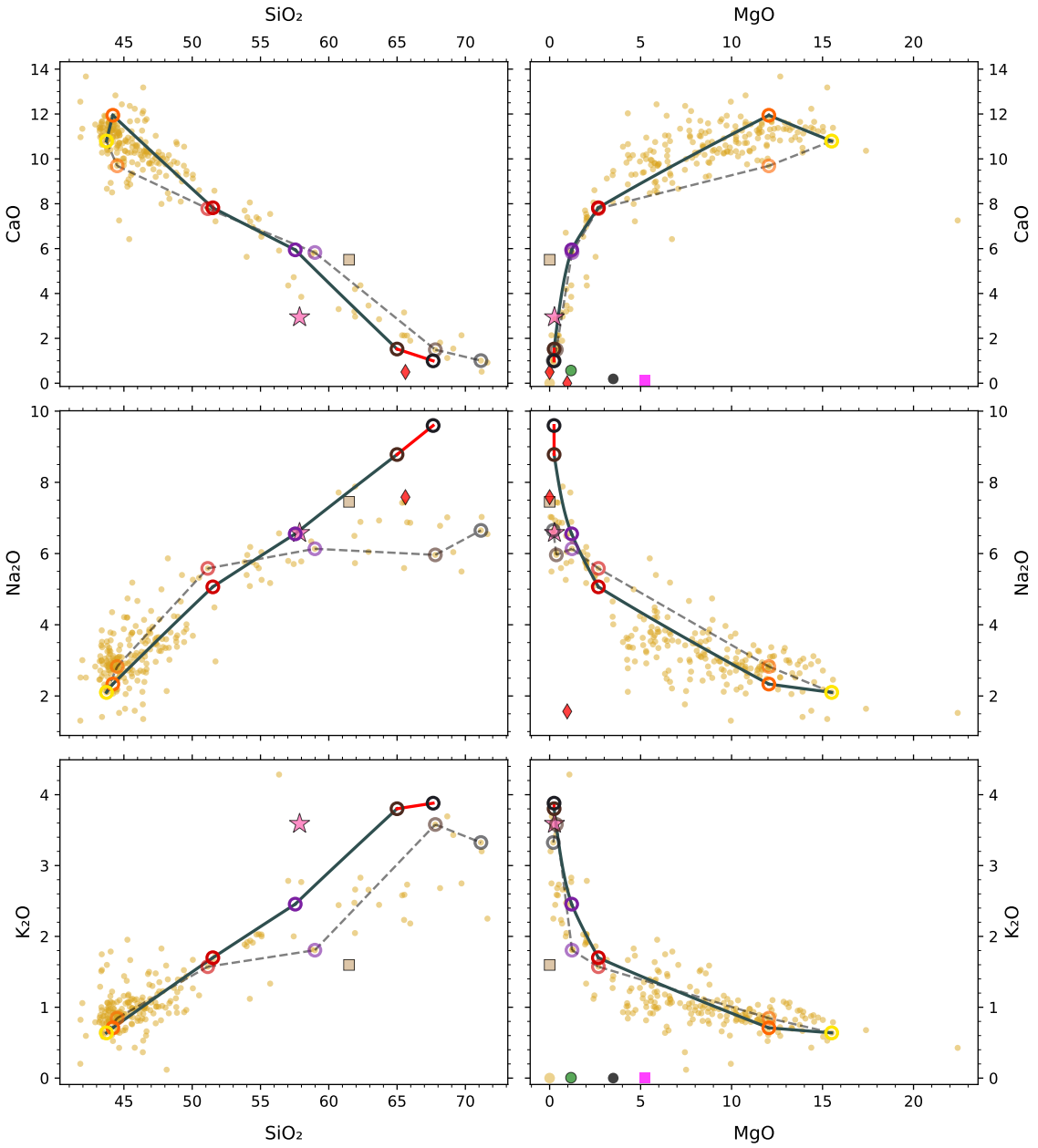



Figure 5: Bivariate plots of the data suite with the data-based and modelled LLODs. The Evolved to End transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

# 5. Figures

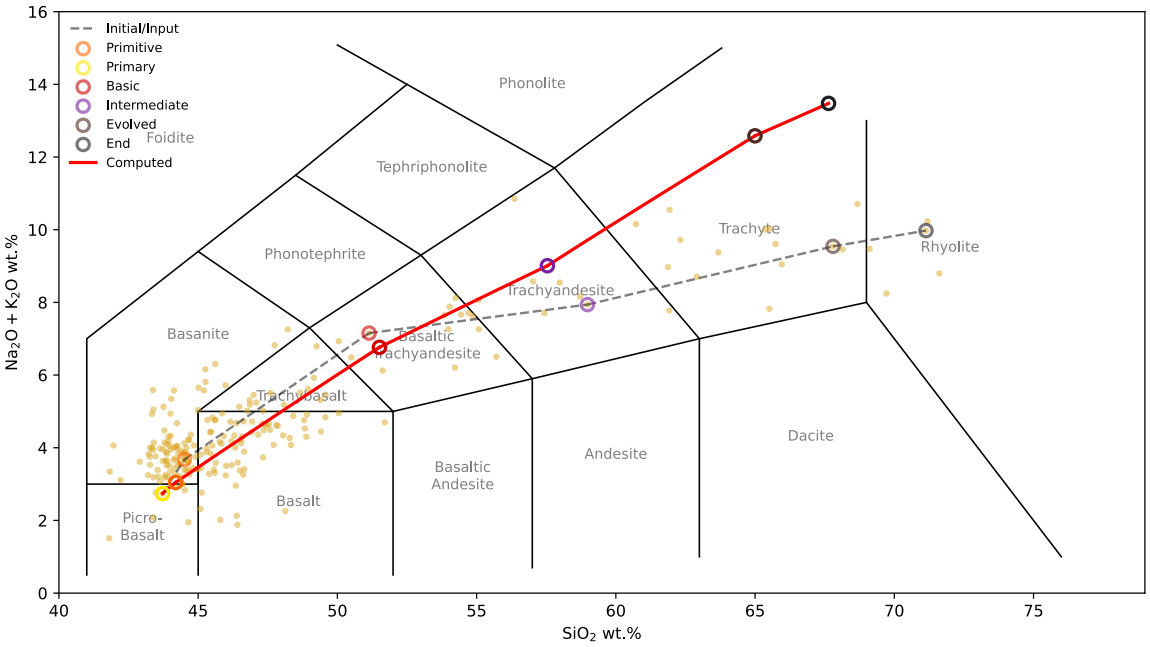



Figure 6: Total Alkali-Silica (TAS) diagram of the model phases.

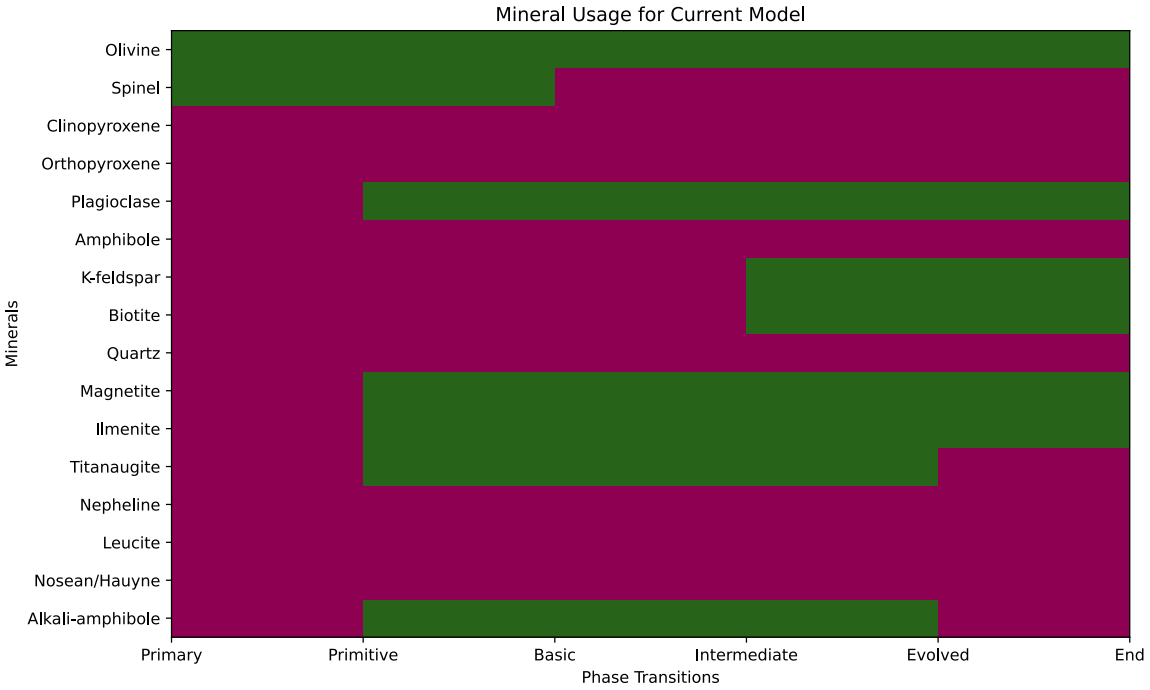



Figure 7: Mineral usage across the model phases.

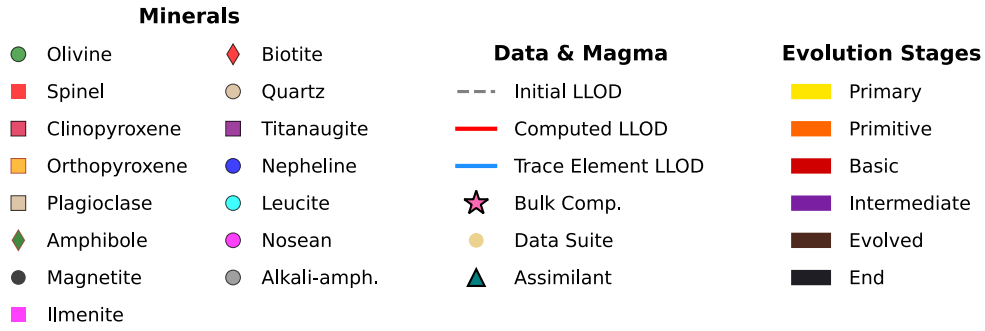



Figure 8: Legend for mineral symbols used in the figures.